

# Methods in Python

22 June 2026 10:59

In Python a method is a function that belongs to a specific object or class. While a regular function is called independently by its name (like `print()`) a method is always invoked on object using dot notation (like `Object.method()`).

```
p = m.exp2(4)
q = m.exp(2)
print(f'e={m.e} to the power of 2 = {q}')
print(p)
```

Output:

```
e=2.718281828459045 to the power of 2 = 7.38905609893065
16.0
```

Method allows object to perform action on their own internal data, forming the backbone of object oriented programming (OOP).

In Python, a method is a function that belongs to an object or a class. Methods are used to perform operations on data associated with that object.

## Difference between function and method

| Function                      | Method                                       |
|-------------------------------|----------------------------------------------|
| Independent block of code.    | Function associated with an object or class. |
| Called directly.              | Called through an Object                     |
| Example: <code>print()</code> | Example: <code>math.exp()</code>             |

## Why use Methods?

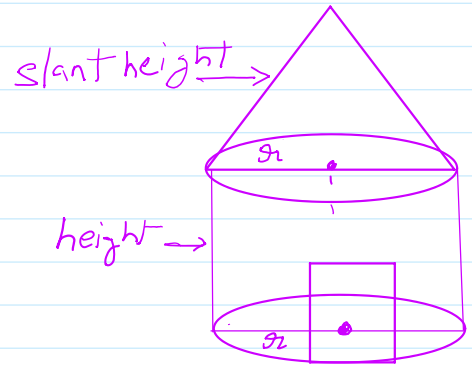
Method helps to:

- Organize code
- Reuse code
- Perform operations on objects
- Implement Object-Oriented Programming(OOP).

In programming, the use of function is one of the means to achieve modularity and reusability. Function can be defined as a named group of instructions that accomplish a specific task when it is invoked. Once defined, a function can be called repeatedly from different places of the program without writing all the code of the that function every time Or it can be called from inside another function by simply writing the name of the function and passing the required parameters if any. The programmer can define as many function as desired while writing the code.

plant height. 

writing the code.



```
# Program to calculate the cost of tent
#
import math
# function definition
def areaOfCyl(h,r):
    area_cyl = 2*math.pi*r*h
    return area_cyl
# Function definition
def areaOfCone(l, r):
    area_con = math.pi*r*l #Area of conical part
    return area_con

# function definition
def post_tax_price(cost):
    tax = 0.18 * cost
    net_price = cost + tax
    return net_price

# main code
h = float(input("Height of cylinder: "))
r = float(input(" Enter radius of cylinder: "))
area_of_cylinder = areaOfCyl(h, r)
l = float(input(" Enter slant height of the conical area in meters : "))
conical_area = areaOfCone(l, r)
canvas_area = conical_area + area_of_cylinder
print(f"Area of canvas: {canvas_area:.2f} m^2")
#calculate the cost of canvas
unit_price = float(input(" Inter cost of 1m square Canvas in rupees: \u20b9 "))
total_cost = unit_price * canvas_area
print(f"The total cost of canvas before tax equal to \u20b9:{total_cost:.2f}")
print(f"Net amount payable(including tax) = \u20b9{post_tax_price(total_cost):.2f}")
```

Output:

```
Height of cylinder: 3
Enter radius of cylinder: 1.5
Enter slant height of the conical area in meters : 1.5
```

Area of canvas: 35.34 m<sup>2</sup>

Inter cost of 1m square Canvas in rupees: ₹ 50

The total cost of canvas before tax equal to ₹:1767.15

Net amount payable(including tax) = ₹ 2085.23

Home work

### 1. Detailed Tent Inventory and Volume Analyzer

Extend the procedural code from the image. Write a program that uses functions to calculate both the surface area and the internal volume of a composite tent structure.

- Create functions `volume_of_cylinder(h, r)` and `volume_of_cone(h_cone, r)`. Derive the vertical height of the cone from the slant height `l` and radius `r` using the math equation: `slant_height_cone = sqrt(l2 - r2)`.
- Print the Canvas Area, Internal Volume, and Total Material Cost.

### 2. Tiered Discount and Multi-State Tax Calculator

In the script, a flat tax of 18% is applied using a simple function. Upgrade this billing flow into a flexible function named `calculate_final_bill(bulk_cost, state_code)`.

- Apply a **tiered discount** to the raw canvas cost before adding tax: 5% discount if the cost is above ₹5,000, 10% if above ₹10,000, and 15% if above ₹20,000.
- Apply different tax rates depending on a string input `state_code` (e.g., "ST1" = 12%, "ST2" = 18%, "ST3" = 28%).

### 3. Tent Structure Waste Material Estimator

When cutting raw canvas sheets to manufacture a tent, a percentage of material is wasted during production.

- Use the exact functions `areaOfCyl(h, r)` and `areaOfCone(l, r)` from the image text.
- Write a main program logic block that prompts the user for a `waste_percentage` integer (e.g., 12% waste).
- Calculate the total raw canvas area that must actually be purchased to account for this production overhead.
- Print the exact monetary cost difference caused strictly by the wasted material using formatted f-strings.